INCH-POUND

MIL-C-83513/2E
31 March 1989
SUPERSEDING
MIL-C-83513/2D
7 March 1986

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, SOCKET CONTACTS, CLASS M, SOLDER TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-83513.

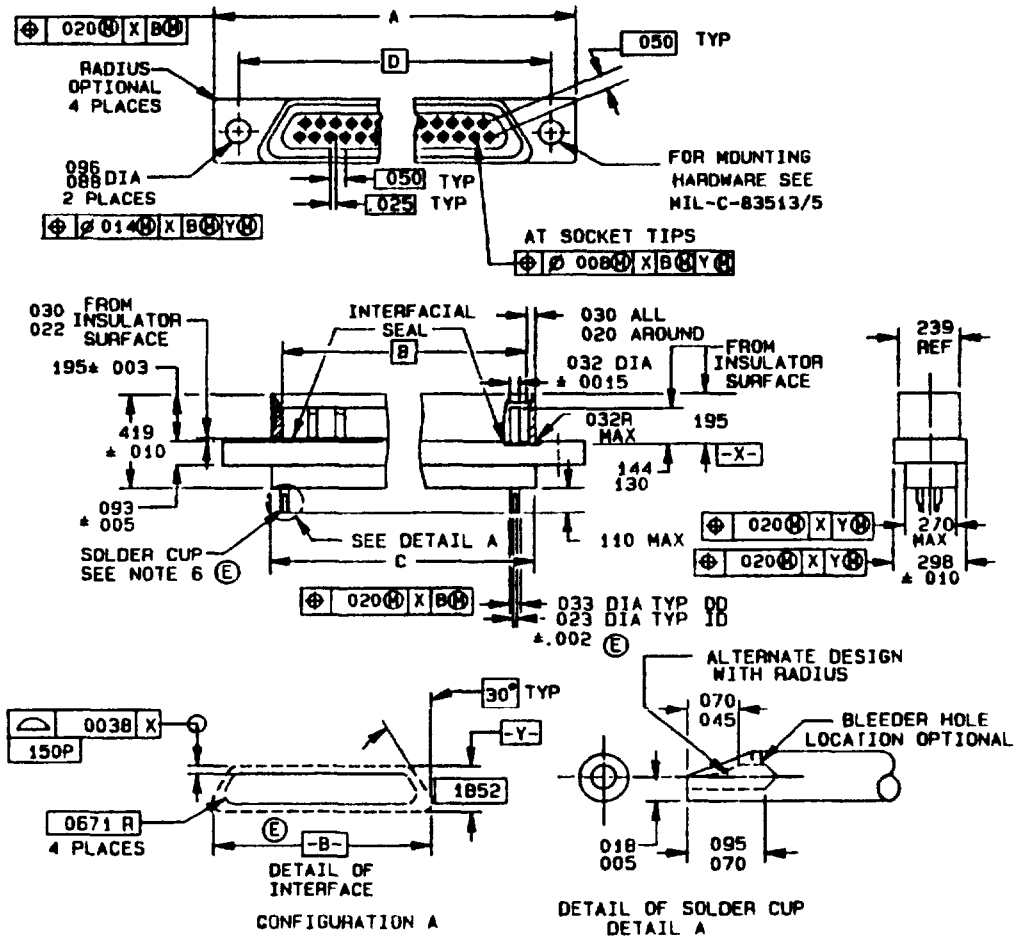


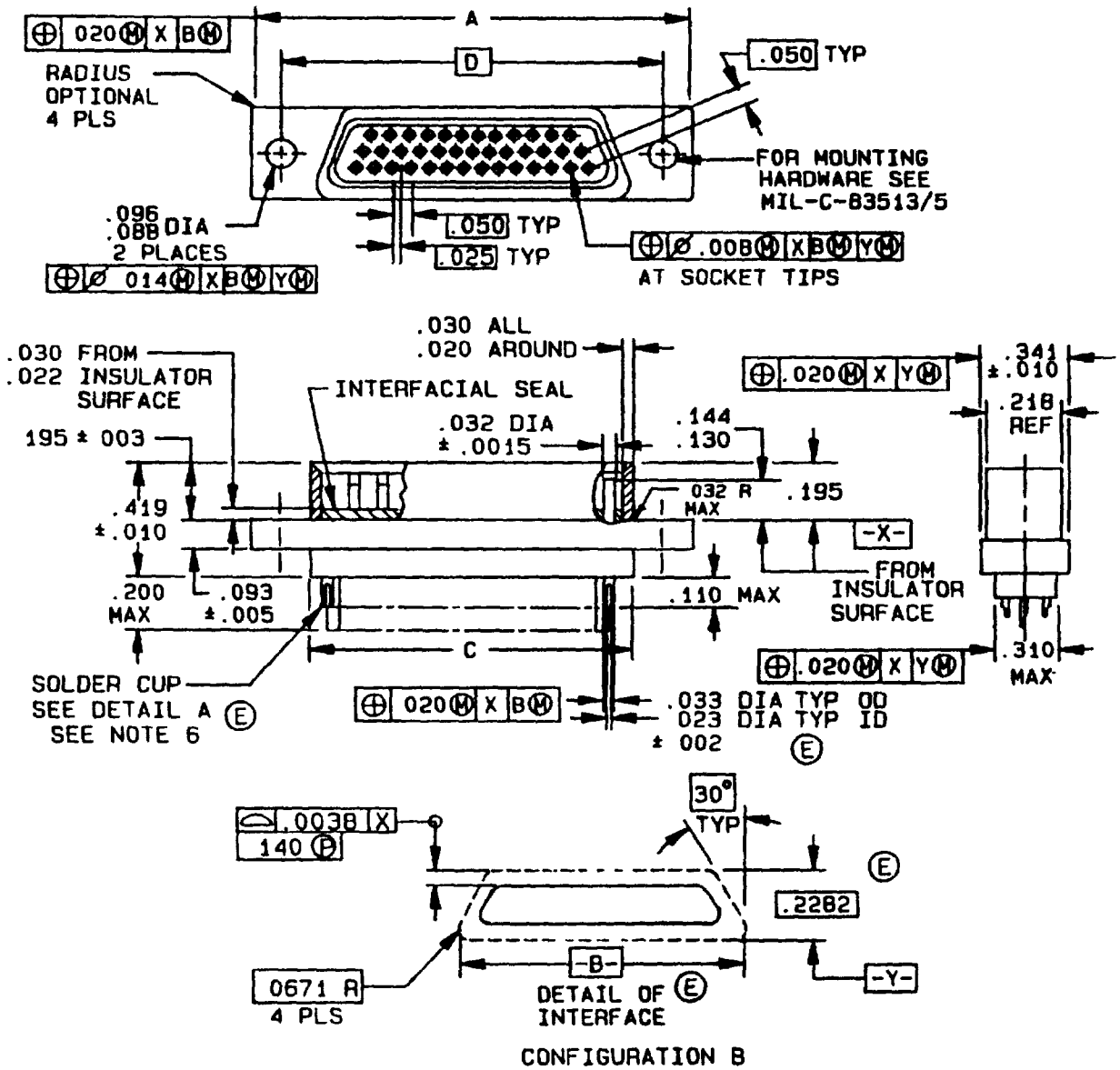
FIGURE 1. Connector, receptacle (.050 spacing).

(E) denotes change

AMSC N/A FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

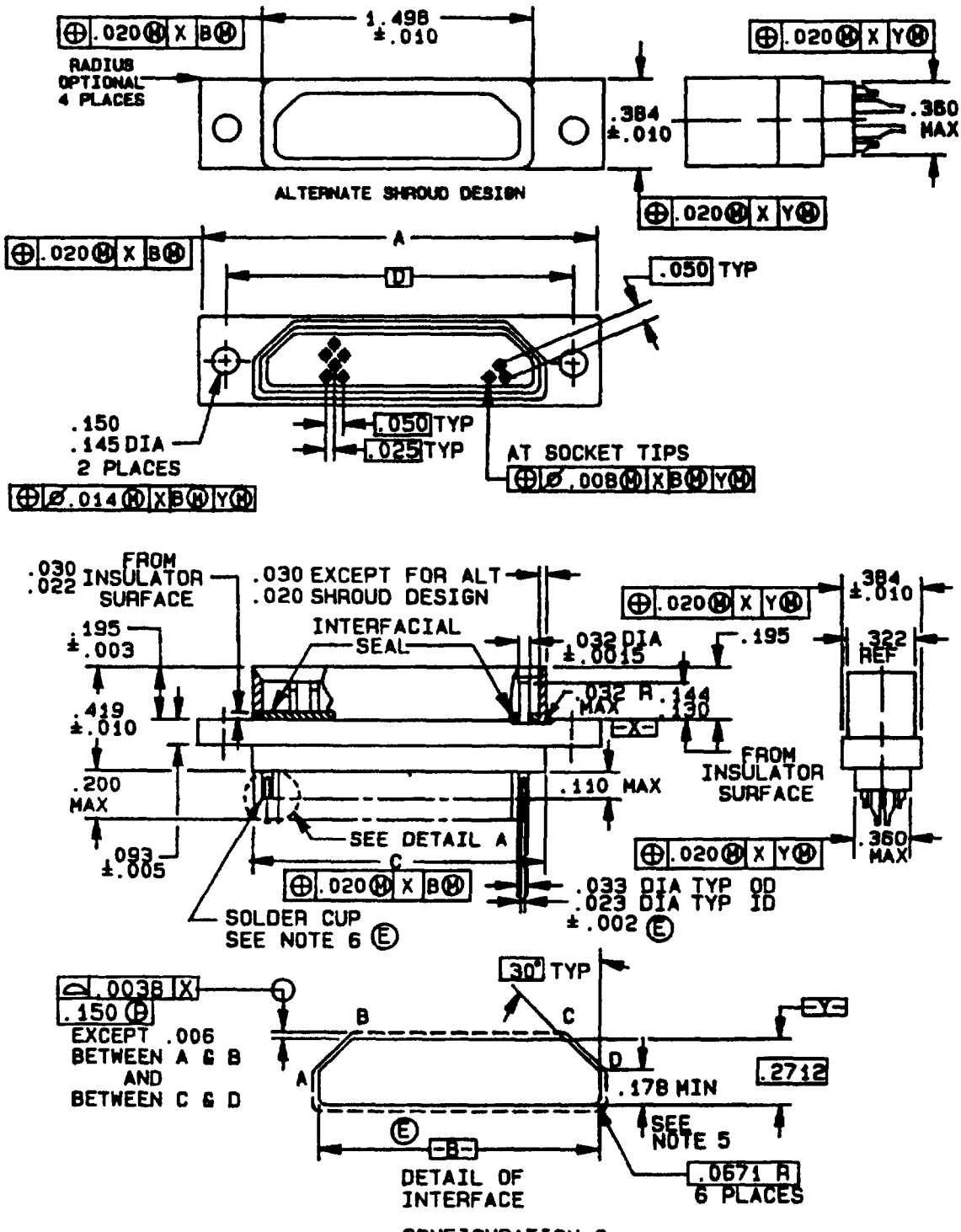
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:27Z
Check the source to verify that this is the current version before use.



CONFIGURATION B

FIGURE 1. Connector, receptacle (.050 spacing) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:27Z
Check the source to verify that this is the current version before use.



CONFIGURATION C

FIGURE 1. Connector, receptacle (.050 spacing) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:27Z
Check the source to verify that this is the current version before use.

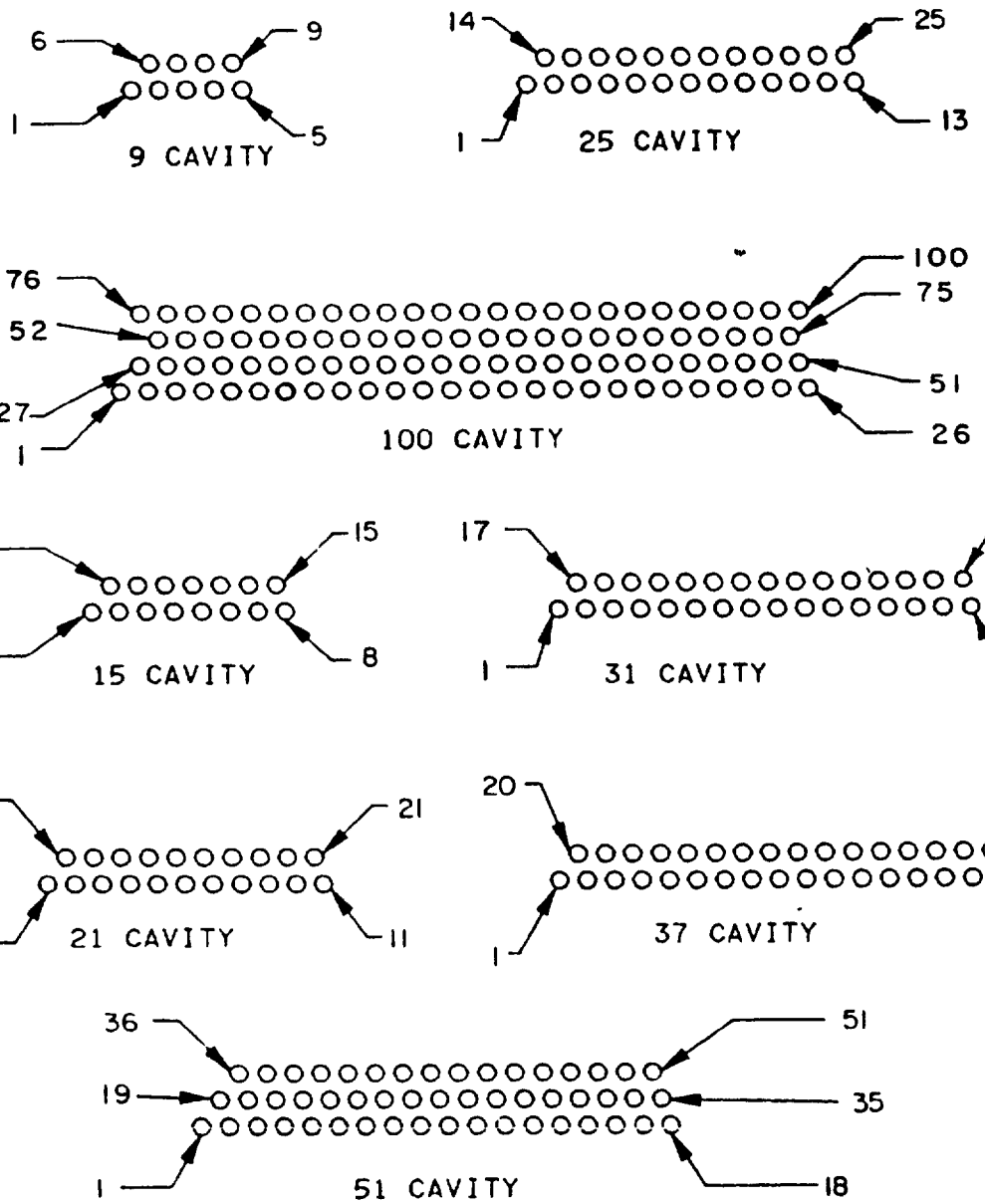
Number cavities	A ±.010 (.25)	B BSC	C ±.010 -.018 +(.25) -(.46)	D BSC	Config. letter
9	.775 (19.68)	.3342 (8.489)	.390 (9.91)	.565 (14.35)	A
15	.925 (23.50)	.4842 (12.299)	.540 (13.72)	.715 (18.16)	A
21	1.075 (27.30)	.6342 (16.109)	.690 (17.53)	.865 (21.97)	A
25	1.175 (29.84)	.7342 (18.649)	.790 (20.07)	.965 (24.51)	A
31	1.325 (33.66)	.8842 (22.459)	.940 (23.88)	1.115 (28.32)	A
37	1.475 (37.46)	1.0342 (26.269)	1.090 (27.69)	1.265 (32.13)	A
51	1.425 (36.20)	.9842 (25.00)	1.040 (26.42)	1.215 (30.86)	B
100	2.160 (54.86)	1.3842 (35.159)	1.432 (36.37)	1.800 (45.72)	C

Inches	mm	Inches	mm
.0015	0.038	.095	2.41
.002	0.05	.100	2.54
.003	0.08	.110	2.79
.0035	0.089	.117	2.97
.0038	0.097	.123	3.12
.004	0.10	.130	3.30
.005	0.13	.144	3.36
.006	0.15	.145	3.68
.007	0.18	.150	3.81
.008	0.20	.180	4.57
.010	0.25	.1852	4.704
.014	0.36	.189	4.80
.015	0.38	.193	4.90
.018	0.46	.195	4.95
.020	0.51	.200	5.08
.022	0.56	.2282	5.796
.025	0.64	.232	5.89
.030	0.76	.239	6.07
.032	0.81	.270	6.86
.033	0.84	.2712	6.888
.045	1.14	.275	6.99
.050	1.27	.281	7.14
.0671	1.704	.298	7.57
.069	1.75	.310	7.87
.070	1.78	.322	8.18
.088	2.24	.341	8.66
.090	2.29	.360	9.14
.0926	2.352	.384	9.75
.093	2.36	.419	10.64
.094	2.39	1.498	37.95

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. Metric equivalents are in parentheses or columns.
5. Dimension line at point D located the point of intersection of the two adjacent side of the receptacle interface.
(E) 6. 26 AWG wire is the maximum wire size that can be used in the solder cup.

FIGURE 1 Connector, receptacle (050 spacing) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:27Z
Check the source to verify that this is the current version before use.



NOTE: Engaging face of socket insert shown.
Cavity identification numbers are for reference only and do not appear on part.

FIGURE 2 Insert arrangement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:27Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1. This specification sheet describes the socket side of a rectangular metal shell connector. This connector uses reverse gender contacts, i.e., the live pin recessed in the insulator with the static socket protruding from a shrouded interface.

Material and finish:
Interfacial seal: Silicone rubber, or fluorosilicone rubber or blend thereof.

Contact identification: See figure 2 for contact layout numbering.

Contact connection: The pin contact (being recessed in the insulator) is normally connected to the live side of the circuit.

Current rating, maximum: 3 amperes per contact.

Mounting and mating hardware: Mounting hardware is to be ordered separately. For insert arrangements A through G, use MIL-C-83513/5, configurations A and B. For insert arrangement H, use MIL-C-83513/5, configuration C (see figure 2).

Mating plug: Shall conform to MIL-C-83513/1 or MIL-C-83513/3.

Ⓔ Part number: Consists of the letter M, the basic number of the specification sheet, a letter from the insert column and the shell finish.

M83513/02 - A C

- M83513/02: Specification sheet number
- A: Insert arrangement (see figure 2)
  - A = 9
  - B = 15
  - C = 21
  - D = 25
  - E = 31
  - F = 37
  - G = 51
  - H = 100
- C: Shell finish
  - C = Cadmium
  - N = Electroless nickel (Space application only)

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, SH
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AT, ME
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-3637-02)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:27Z
Check the source to verify that this is the current version before use.